

Documentation chemical data input

This document contains the documentation for the chemical data input in the program “**interfaz_remix**”. There are maximum 5 files to be read. The minimum of input files is 3 (the ones that end in “_quim_loc.dat”, “_sist_quim.dat” and “_tar_wat.dat”). The other two (ending in “_tar_sol.dat” and “_tar_gas.dat”) are optional, depending if the chemical system contains solids and/or gases. All of the files must contain the **same root** in their names, and they must live in the **same directory**.

We describe the input files with an example of two waters that mix: an inflow water in equilibrium with atmospheric O_2 and CO_2 and a resident water in equilibrium with calcite. This example can be found in the directory “..\examples\calcite_eq” in the repository.

Chemical system file: “_sist_quim.dat”

This file contains the information related to the chemical system (species, reactions, etc). It is read by the **Chemical System** class.

- **Line 1:** Title of the problem
- **Line 2:** Section separation
- **Line 3:** Section name
 - If Section name = ‘PRIMARY AQUEOUS SPECIES’ or ‘AQUEOUS COMPLEXES’:
 - i. **Line 3a:** *sp_name* (string)
 - i. Name of species in database “*master25_modif.dat*”
 - ii. **Line 3b:** end of section
 - If Section name = ‘MINERALS’ or ‘GASES’:
 - i. **Line 3a:** *sp_name* (string), *flag_eq* (logical), *flag_cst_act* (logical)
 - i. *sp_name*: Name of species in database “*master25_modif.dat*”
 - ii. *flag_eq*: TRUE if equilibrium reaction, FALSE otherwise
 - iii. *flag_cst_act*: TRUE if species has constant activity, FALSE otherwise
 - ii. **Line 3b:** end of section
 - If Section name = ‘REDOX REACTIONS’:
 - i. **Line 3a:** *react_name* (string), *flag_eq* (logical)
 - i. *react_name*: Name of redox reaction in database “*reacciones_monod.dat*”
 - ii. *flag_eq*: TRUE if equilibrium reaction, FALSE otherwise
 - ii. **Line 3b:** end of section
 - If Section name = ‘SURFACE COMPLEXES’:
 - i. **Line 3a:** *sp_name* (string)
 - i. *sp_name*: Name of species in database “*master25_modif.dat*”
 - ii. **Line 3b:** end of section
- **Line 4:** Section separation
- **Line 5:** end of file

'TITLE OF THE PROBLEM: Calcite in equilibrium'

'-----'

'PRIMARY AQUEOUS SPECIES'

'h2o(p)' ! name of primary aqueous species

'h+'

'ca+2'

'o2(aq)'

'hco3-'

'*'

'MINERALS'

'calcite' .true. .true. ! name of mineral, flag equilibrium, flag constant activity

'*'.true. .true.

'GASES'

'o2(g)' .true. .true. ! name of gas, flag equilibrium, flag constant activity

'co2(g)' .true. .true. ! name of gas, flag equilibrium, flag constant activity

'*'.true. .true.

'-----'

'end'

Local chemistry file: “_quim_loc.dat”

This file contains the **local** chemical information (water types, mineral zones, gas zones). It is read by the **Chemistry** class.

- **Line 1:** Name of the problem
- **Line 2:** Section separation
- **Section name:** ‘INITIAL AND BOUNDARY WATER TYPES’
 - **Line 1:** *act_coeffs_model* (integer)
 - 0. Ideal
 - 1. Debye-Hückel
 - 2. Debye-Hückel extended
 - 3. Davies
 - 4. Truesdell-Jones
 - **Line 2:** *num_wat_types* (integer)
 - i. Number of water types (includes initial, recharge, boundary, etc)
 - **Line 3:** *ind_wat_type* (integer), *temp* (real)
 - i. *ind_wat_type*: index of water type
 - ii. *temp*: temperature of solution (in Celsius)
 - **Line 3a:** *name* (string)
 - i. Name of water type
 - **Line 3b:** water type data begins
 - i. *icon*: initial condition type
 - 1. prescribed primary concentration
 - 2. prescribed total concentration
 - 3. prescribed activity
 - 4. primary species is in equilibrium with a phase
 - ii. *guess*: Initial guess for primary species concentration
 - iii. *ctot*: (all concentrations are expressed in molalities)
 - i. *icon*=1: primary species concentration
 - ii. *icon*=2: total concentration of primary species
 - iii. *icon*=3: activity of primary species
 - iv. *icon*=4: activity of phase that is in equilibrium with this primary species
 - iv. *constrain*: if *icon*=4, it is the phase in equilibrium with the primary species
 - **Line 3c:** *prim_sp_name* (string), *icon* (integer), *guess* (real), *ctot* (real), *constrain* (string)
 - **Line 3d:** end of water type data
 - **Line 4:** end of section

'TITLE OF THE PROBLEM: Calcite in equilibrium'

'-----'

'INITIAL AND BOUNDARY WATER TYPES'

0 ! activity coefficients model

2 ! number of water types

1 25.0 ! index water type, temperature (C)

'boundary' ! name of water type

' icon guess ctot constrain'

'h2o(p)' 3 1d0 1d0 "

'ca+2' 1 1d-9 1d-9 "

'o2(aq)' 4 1d-9 2d-1 'o2(g)'

'hco3-' 4 1d-9 3.5d-4 'co2(g)'

'h+' 1 1d-7 1d-7 "

'*' 0 0.0 0.0 "

2 25.0 ! index water type, temperature (C)

'resident' ! name of water type

' icon guess ctot constrain'

'h2o(p)' 3 1d0 1d0 "

'ca+2' 1 1d-3 1d-3 "

'o2(aq)' 1 1d-6 1d-6 "

'hco3-' 4 1d-9 1d0 'calcite'

'h+' 1 1d-5 1d-5 "

'*' 0 0.0 0.0 "

'-----'

- Section name: 'INITIAL MINERAL ZONES'
 - **Line 1:** *num_min_zones* (integer)
 - i. Number of mineral zones
 - **Line 2a:** *ind_min_zone* (integer), *temp* (real)
 - i. *ind_min_zone*: index of mineral zone
 - ii. *temp*: temperature of mineral zone (in Celsius)
 - **Line 2b:** *name* (string)
 - i. Name of mineral zone
 - **Line 2c:** mineral zone data begins
 - **Line 2d:** *min_name* (string), *vol_frac* (real), *react_surf* (real)
 - i. *min_name*: name of mineral in mineral zone (must be in the chemical system)
 - ii. *vol_frac.*: initial volumetric fraction of mineral
 - iii. *react_surf*: specific surface of mineral (surface of mineral per unit volume of rock)
 - **Line 2e:** end of mineral zone data
 - **Line 3:** end of section

'INITIAL MINERAL ZONES'

1 ! number of mineral zones

1 25.0 ! index of mineral zone, temperature (C)

'calcite' ! name of mineral zone

'mineral vol.frac. area'

'calcite' 0.50 1d2

'*' 0.00 0.00

'-----'

- Section name: 'INITIAL AND BOUNDARY GAS ZONES'
 - **Line 1:** *num_gas_zones* (integer)
 - i. Number of gas zones
 - **Line 2a:** *ind_gas_zone* (integer), *temp* (real), *vol* (real)
 - i. *ind_gas_zone*: index of gas zone
 - ii. *temp*: temperature of gas zone (in Celsius)
 - iii. *vol*: total volume of gas
 - **Line 2b:** gas zone data begins
 - **Line 2c:** *name* (string)
 - i. Name of gas zone
 - **Line 2d:** *gas_name* (string), *part_press* (real)
 - i. *gas_name*: name of gas in gas zone (must be in the chemical system)
 - ii. *part_press*: initial partial pressure (equivalent to activity) of gas (in atm)
 - **Line 2e:** end of gas zone data
 - **Line 3:** end of section
 - **Line 4:** end of file

'INITIAL AND BOUNDARY GAS ZONES'

1 ! number of gas zones

1 25 1d9 ! index gas zone, temperature (C), volume

'atmosphere' ! name of gas zone

'gas partial pressure'

'o2(g)' 2d-1

'co2(g)' 3.5d-4

'*' 0.00

'-----'

'end'

Target waters file: “_tar_wat.dat”

This file contains the **association** between target waters, target solids and/or target gases, and it also indicates if the target waters are external or not. This file is read by the **Chemistry** class.

- **Line 1:** Name of the problem
- **Line 2:** Section separation
- **Section name:** 'TARGET WATERS'
 - **Line 1:** *num_tar_wat* (integer)
 - Number of target waters
 - **Line 2:** *num_rech_wat* (integer)
 - Number of recharge target waters
 - **Line 3:** *num_bd_wat* (integer)
 - Number of boundary target waters
 - **Lines 4 - 4+n^o intervals:** *int_wat* (string), *iwtype* (integer), *int_sol* (integer), *int_gas*(integer), *flag_wat_type*(logical)
 - *int_wat*: Interval of target waters
 - *iwtype*: Index water type in section 'INITIAL AND BOUNDARY WATER TYPES'
 - *int_sol*: Interval of target solids associated with *int_wat*
 - *int_gas*: Interval of target gases associated with *int_wat*
 - *flag_wat_type*: Flag to indicate target water type.
 - 0: Boundary
 - 1: Domain
 - 2: Recharge
 - End of section

'TITLE OF THE PROBLEM: Calcite in equilibrium'

'TARGET WATERS'

12 ! number of target waters

0 ! number of recharge target waters

2 ! number of boundary target waters

1	1	1	1	0	! target waters interval, water type index, target solids interval, target gases interval, flag water type
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2-11 2 1-10 0 1

12 2 10 0 0

'end'

Target solids file: “_tar_sol.dat”

This file contains the **association** between target solids and solid zones. It is read by the **Chemistry** class. The only solid zones implemented so far are the **mineral** zones. **IMPORTANT:** this file will not be read by the code if there are no solid zones.

- **Line 1:** Name of the problem
- **Line 2:** Section separation
- **Section name:** ‘TARGET SOLIDS’
 - **Line 1:** *num_tar_sol* (integer)
 - Number of target solids
 - **Line 2:** *interval* (string), *iszone* (integer)
 - *interval*: Target solids interval
 - *iszone*: Index of mineral zone in section ‘INITIAL MINERAL ZONES’
 - end of section

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'TITLE OF THE PROBLEM: Calcite in equilibrium'
'-----'
'TARGET SOLIDS'

10                                ! number of target solids

1-10    1                        ! target solids interval, solid zone index
'-----'

'end'
```

Target gases file: “_tar_gas.dat”

This file contains the **association** between target gases and gas zones. It is read by the **Chemistry** class. **IMPORTANT:** this file will not be read by the code if there are no gas zones.

- **Line 1:** Name of the problem
- **Line 2:** Section separation
- **Section name:** ‘TARGET GASES’
 - **Line 1:** *num_tar_gas* (integer)
 - Number of target gases
 - **Line 2:** *interval* (string), *igzone* (integer)
 - *interval*: Target solids interval
 - *igzone*: Index of gas zone in section ‘INITIAL AND BOUNDARY GAS ZONES’
 - end of section

```
'TITLE OF THE PROBLEM: Calcite in equilibrium'
'-----'
'TARGET GASES'

1                                ! number of target gases

1    1                          ! target gases interval, gas zone index
'-----'

'end'
```